A Lamport-Style Proof Package for the Lower Bound $g(n)\geq \lceil 7n/27\rceil$

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Let G be the Euclidean minimum-distance graph of a finite point set $P\subset\mathbb{R}^2$: all distinct points of P have distance at least 1, and two vertices are adjacent exactly when their distance is 1. Let $g(n)$ be the minimum possible value of the independence number among such graphs on n vertices. This manuscript proves, relative to the published broken-lattice input from Swanepoel's proof of the $8/31$ bound, that

$$\alpha(G)\geq \frac{7}{27}|V(G)|,$$

$$g(n)\geq \left\lceil \frac{7n}{27}\right\rceil .$$

The proof is a minimal-counterexample argument. Swanepoel's broken-lattice machinery is specialized to $m=7$. The only possible equality break is then $s_3\neq r_4$. The remaining local configurations are eliminated by explicit independent-set ledgers. All neighbourhoods in the proof are open neighbourhoods.

Definitions and external input

Definition — Euclidean minimum-distance graph

A Euclidean minimum-distance graph is a graph G obtained from a finite point set $P\subset\mathbb{R}^2$ satisfying

$$|x-y|\geq 1 \quad (x\neq y,\ x,y\in P),$$

with vertex set P and an edge xy exactly when

$$|x-y|=1.$$

For $S\subseteq V(G)$, write

$$N(S)=\{v\in V(G)\setminus S: v \text{ is adjacent to some } s\in S\}.$$

Definition

Let $g(n)$ be the minimum of $\alpha(G)$ over all Euclidean minimum-distance graphs G on n vertices.

Proposition — Swanepoel broken-lattice input

We use the following published facts from Swanepoel's proof of the lower bound for planar contact graphs [Swanepoel2002]. The boundary-arc structure is Theorem 4 on p. 661; the two-extra-neighbour statement is Lemma 8 on pp. 659--660; the triple obstruction and the cap used below come from Lemma 9 and its proof on pp. 660--661; and the Euclidean all-but-one equality and break inequality are Lemma 10 and inequality (5) on pp. 661--665.

Let $m\geq 5$, and let G be a smallest counterexample to

$$\alpha(G)\geq \frac{m}{4m-1}|V(G)|.$$

Then there is a nonconcave boundary arc

$$p_0,p_1,\dots,p_{2m-3}$$

with total turn less than $\pi/3$, with

$$\deg(p_0)=3,$$

$$\deg(p_i)=4 \quad (1\leq i\leq 2m-3),$$

and with a unique common neighbour q_i of p_i and p_{i+1} for each boundary edge p_ip_{i+1}.

For each

$$1\leq i\leq 2m-5,$$

Swanepoel's Lemma 8, pp. 659--660, gives exactly two extra neighbours r_i,s_i of q_i outside the local set

$$\{p_i,p_{i+1},q_{i-1},q_{i+1}\},$$

where unavailable endpoint q-terms are omitted.

If

$$s_i=r_{i+1} \quad (i=a,a+1,a+2),$$

then Swanepoel's Lemma 9, pp. 660--661, gives

$$a\ge 2m-10.$$

Moreover, the proof of Lemma 9 on pp. 660--661 gives the cap

$$|N(s_{a+1})\setminus\{s_a,s_{a+2},q_{a+1},q_{a+2}\}|\le 2.$$

In the Euclidean plane, Swanepoel's Lemma 10 on pp. 661--665 gives

$$s_i=r_{i+1} \quad (1\le i\le 2m-6)$$

for all but at most one index i. If

$$s_i\ne r_{i+1},$$

then Lemma 10, specifically inequality (5), gives the break inequality

$$\tau_i+2\tau_{i+1}+\tau_{i+2}\ge \frac{\pi}{3}.$$

Remark

The proof below proves the theorem relative to Proposition prop:SwanepoelInput. The endpoint use of q_1 and q_9 uses Lemma 8 in the range $1\le i\le 2m-5$. The cap for t_5,t_6,t_7 is the cap obtained in the proof of Lemma 9, not a separate theorem statement. The uniform side formula for q_i is not imported from Swanepoel; it is proved internally in Lemma lem:boundary-side.

The deletion criterion

Lemma — Deletion criterion

Let G be a smallest counterexample to

$$\alpha(G)\ge \frac{7}{27}|V(G)|.$$

If $S\subseteq V(G)$ is independent and

$$|N(S)|\le \left\lfloor\frac{20|S|}{7}\right\rfloor,$$

then G is not a counterexample.

Proof.

Let

$$n=|V(G)|.$$

Set

$$G'=G-(S\cup N(S)).$$

By minimality of G,

$$\alpha(G')\ge \frac{7}{27}|V(G')|.$$

Since no vertex of G' is adjacent to S, an independent set in G' can be joined with S. Therefore

$$\alpha(G)\ge |S|+\alpha(G').$$

Also

$$|V(G')|=n-|S|-|N(S)|.$$

Hence

$$\begin{aligned}
\alpha(G)
&\ge |S|+\frac{7}{27}(n-|S|-|N(S)|)\\
&=\frac{7n}{27}+\frac{20|S|-7|N(S)|}{27}.
\end{aligned}$$

If $|N(S)|\le 20|S|/7$, then the last numerator is nonnegative. Thus

$$\alpha(G)\ge \frac{7n}{27},$$

contradicting that G is a counterexample.

The $m=7$ broken-lattice structure

Assume, for contradiction, that G is a smallest counterexample to

$$\alpha(G)\geq \frac{7}{27}|V(G)|.$$

Apply Proposition prop:SwanepoelInput with $m=7$. Then

$$2m-3=11,$$

$$2m-6=8,$$

$$2m-10=4.$$

Thus the boundary arc is

$$p_0,p_1,\dots,p_{11}.$$

Lemma — The unique $m=7$ break

In this $m=7$ broken lattice, the only possible failed equality among

$$s_i=r_{i+1} \quad (1\leq i\leq 8)$$

is

$$s_3\neq r_4.$$

Consequently,

$$s_i=r_{i+1} \quad i=1,2,4,5,6,7,8.$$

Proof.

By Proposition prop:SwanepoelInput, at most one equality among

$$s_i=r_{i+1} \quad (1\leq i\leq 8)$$

fails. If three consecutive equalities begin at a, then $a\geq 4$. Therefore true triples beginning at $a=1,2,3$ are impossible. These forbidden triples are

$$(1,2,3), \quad (2,3,4), \quad (3,4,5).$$

A single failed equality must meet all three triples. Their only common index is 3. Hence the sole possible break is

$$s_3\neq r_4.$$

Whenever $s_i=r_{i+1}$, write

$$t_i=s_i=r_{i+1}.$$

Thus t_i is defined for

$$i=1,2,4,5,6,7,8.$$

Boundary-side orientation

Lemma — Straight-line planarity

The straight-line drawing of a Euclidean minimum-distance graph has no crossing pair of edges.

Proof.

Suppose two unit edges ab and cd cross at an interior point x. One endpoint of ab, say u, satisfies

$$|u-x|\leq \frac12,$$

and one endpoint of cd, say v, satisfies

$$|v-x|\leq \frac12.$$

The edges are distinct and cross properly, so $u\neq v$. If equality held in

$$|u-v|\leq |u-x|+|x-v|\leq 1,$$

then u,x,v would be collinear and both distances would be exactly $1/2$, which is incompatible with two distinct straight segments crossing properly. Hence

$$|u-v|<1,$$

contradicting the minimum-distance hypothesis.

Lemma — Empty unit triangle

Let abc be a unit equilateral triangle whose vertices are in the point set. No other vertex of the point set lies in the closed triangular region bounded by abc.

Proof.

The closed triangular region has diameter 1. If a distinct point z lay in it, then the minimum-distance condition would require

$$|z-a|\ge 1,\quad |z-b|\ge 1,\quad |z-c|\ge 1.$$

Since the region has diameter 1, all three distances would have to be exactly 1. The only point in the closed triangle at distance 1 from both a and b is c. Thus no distinct point z exists.

Lemma — Boundary-side orientation

Let p_ip_{i+1} be one of the boundary triangle edges in the Swanepoel arc, and let q_i be the unique common neighbour of p_i and p_{i+1}. Orient the boundary cycle so that the interior of the drawing is locally on the left of each directed boundary edge in the arc. Then q_i lies on the interior side of the directed edge p_ip_{i+1}. Consequently, after choosing the complex coordinate orientation so that left rotation by 60° is multiplication by $\omega=e^{i\pi/3}$,

$$q_i=p_i+\omega(p_{i+1}-p_i)$$

for every relevant i in the arc.

Proof.

Write $a=p_i$, $b=p_{i+1}$, and $q=q_i$. Since q is adjacent to both endpoints of the unit edge ab,

$$|a-b|=|a-q|=|b-q|=1.$$

Thus a,b,q form a unit equilateral triangle, and q is one of the two equilateral apices on the segment ab.

The segment ab is an outer-boundary edge of the straight-line plane drawing. One side of ab is incident with the outer face, and the other side is the interior side of the drawing. Suppose, for contradiction, that q lies on the outer-face side of ab. Then the triangular cycle $a\ q\ b\ a$ lies on the outer-face side of ab. By Lemma lem:empty-triangle, its closed triangular region contains no other graph vertex. By Lemma lem:straight-planar, no graph edge crosses its boundary. Therefore the bounded triangular region enclosed by a,q,b is a face of the plane embedding. On that side the edge ab is then incident with this bounded triangular face, not with the outer face. This contradicts that ab is an outer-boundary edge with the outer face on that side.

Hence q lies on the interior side of ab. By the chosen orientation of the boundary, the interior side is the left side of the directed edge ab. The left equilateral apex of the directed unit segment from a to b is

$$a+\omega(b-a).$$

Thus

$$q_i=p_i+\omega(p_{i+1}-p_i).$$

Coordinate lemmas

Throughout, identify $\mathbb{R}^2$ with $\mathbb{C}$ according to Lemma lem:boundary-side, and put

$$\omega=e^{i\pi/3}.$$

Write

$$e_i=p_{i+1}-p_i,\quad |e_i|=1.$$

By Lemma lem:boundary-side,

$$q_i=p_i+\omega e_i.$$

Let τ_{i+1} denote the clockwise turn from e_i to e_{i+1}, so

$$e_{i+1}=e_i e^{-i\tau_{i+1}}, \qquad \tau_{i+1}\ge 0.$$

The total turn of the arc is less than $\pi/3$. This convention is local: only turn sums involving existing consecutive edges are used below.

Lemma — Degree bound

Every vertex of a Euclidean minimum-distance graph has degree at most 6.

Proof.

Let v be a vertex. Its neighbours lie on the unit circle centred at v. If x,y are two distinct neighbours and θ is their smaller angular separation around v, then

$$|x-y|=2\sin(\theta/2).$$

Since all distinct vertices are at distance at least 1,

$$2\sin(\theta/2)\ge 1,$$

and hence

$$\theta\ge \pi/3.$$

At most six directions around a circle can be pairwise separated by at least $\pi/3$. Therefore $\deg(v)\le 6$.

Lemma — General q_iq_{i+1} turn criterion

For every relevant i,

$$q_iq_{i+1}\in E(G) \Longleftrightarrow \tau_{i+1}=0.$$

This statement does not assume $s_i=r_{i+1}$.

Proof.

Using $q_i=p_i+\omega e_i$,

$$\begin{aligned} q_{i+1}-q_i &=p_{i+1}+\omega e_{i+1}-p_i-\omega e_i\\ &=e_i+\omega e_{i+1}-\omega e_i\\ &=(1-\omega)e_i+\omega e_{i+1}\\ &=\bar\omega e_i+\omega e_{i+1}. \end{aligned}$$

The two summands are unit vectors. Since e_{i+1} is obtained from e_i by a clockwise turn τ_{i+1}, the angle between $\bar\omega e_i$ and ωe_{i+1} is

$$\frac{2\pi}{3}-\tau_{i+1}.$$

Because $0\le \tau_{i+1}<\pi/3$, this angle lies in $(\pi/3,2\pi/3]$. A sum of two unit vectors has length 1 exactly when the angle between them is $2\pi/3$. Therefore

$$|q_{i+1}-q_i|=1 \Longleftrightarrow \tau_{i+1}=0.$$

Since edges are exactly unit distances, the result follows.

Lemma — Equality-strip formula and t_it_{i+1} criterion

If $s_i=r_{i+1}$, then

$$t_i=p_i+\omega(e_i+e_{i+1}).$$

If in addition both t_i and t_{i+1} are defined, then

$$t_it_{i+1}\in E(G) \Longleftrightarrow \tau_{i+1}+\tau_{i+2}=0.$$

Proof.

The two unit-circle intersections of q_i and q_{i+1} are p_{i+1} and

$$q_i+q_{i+1}-p_{i+1}.$$

On an equality strip this second common neighbour is t_i. Thus

$$\begin{aligned}
t_i&=q_i+q_{i+1}-p_{i+1}\\
&=(p_i+\omega e_i)+(p_{i+1}+\omega e_{i+1})-p_{i+1}\\
&=p_i+\omega(e_i+e_{i+1}).
\end{aligned}$$

Now suppose t_i,t_{i+1} are defined. Then

$$\begin{aligned}
t_{i+1}-t_i
&=p_{i+1}+\omega(e_{i+1}+e_{i+2})-p_i-\omega(e_i+e_{i+1})\\
&=e_i-\omega e_i+\omega e_{i+2}\\
&=\bar\omega e_i+\omega e_{i+2}.
\end{aligned}$$

The angle between these two unit summands is

$$\frac{2\pi}{3}-(\tau_{i+1}+\tau_{i+2}).$$

Thus the length is 1 exactly when this angle is $2\pi/3$, equivalently when

$$\tau_{i+1}+\tau_{i+2}=0.$$

Corollary

If $t_4t_5\in E(G)$, then

$$\tau_5=\tau_6=0.$$

After this, the following cases are exhaustive:

$$\tau_7>0,$$

$$\tau_7=0,\quad \tau_8+\tau_9>0,$$

and

$$\tau_7=\tau_8=\tau_9=0.$$

Proof.

By Lemma lem:t-equality,

$$t_4t_5\in E(G) \Longleftrightarrow \tau_5+\tau_6=0.$$

Since both turns are nonnegative, this is equivalent to $\tau_5=\tau_6=0$. The trichotomy is then the exhaustive trichotomy for the later nonnegative turns τ_7,τ_8,τ_9.

Lemma — Forward cone estimate

Rotate so that $e_a=1$. For every $j\geq a$, there exists $\phi_j\in[0,\pi/3)$ such that

$$e_j=e^{-i\phi_j}.$$

Consequently,

$$\mathrm{Re}(e_j)>\frac12,$$

$$\mathrm{Re}(\omega e_j)\geq\frac12.$$

Proof.

All turns are nonnegative and the total turn is less than $\pi/3$. Hence every later direction is obtained from e_a by a clockwise turn $\phi_j\in[0,\pi/3)$. Therefore

$$e_j=e^{-i\phi_j}.$$

Thus

$$\mathrm{Re}(e_j)=\cos\phi_j>\frac12.$$

Also

$$\omega e_j=e^{i(\pi/3-\phi_j)},$$

and $0\leq \pi/3-\phi_j\leq \pi/3$, giving

$$\mathrm{Re}(\omega e_j)\geq\frac12.$$

Lemma — Triangular-lattice unit vectors

Let

$$z=a+b\omega,\qquad a,b\in\mathbb Z.$$

Then

$$|z|^2=a^2+ab+b^2.$$

Moreover, $|z|=1$ if and only if

$$z\in\{\pm1,\pm\omega,\pm\omega^2\}.$$

Proof.

Since $\omega+\bar\omega=1$,

$$\begin{aligned}
|a+b\omega|^2
&=(a+b\omega)(a+b\bar\omega)\\
&=a^2+ab(\omega+\bar\omega)+b^2\\
&=a^2+ab+b^2.
\end{aligned}$$

If $a^2+ab+b^2=1$, then

$$a^2+ab+b^2=(a+\frac b2)^2+\frac34 b^2.$$

Thus $|b|\le1$. If $b=0$, then $a=\pm1$. If $b=1$, then $a^2+a=0$, so $a=0$ or $a=-1$, giving ω or ω^2. If $b=-1$, then $a^2-a=0$, so $a=0$ or $a=1$, giving $-\omega$ or $-\omega^2$. This gives exactly the six listed units.

Boundary, endpoint, and separation lemmas

Lemma — Boundary neighbourhoods

For $1\le i\le 10$,

$$N(p_i)=\{p_{i-1},p_{i+1},q_{i-1},q_i\}.$$

Also, for a vertex p_{-1},

$$N(p_0)=\{p_{-1},p_1,q_0\},$$

and

$$N(p_{11})\subseteq \{p_{10},q_{10}\}\cup U_{11},\qquad |U_{11}|\le2.$$

Proof.

For $1\le i\le10$, Swanepoel gives $\deg(p_i)=4$. The four displayed vertices are neighbours of p_i. We verify that they are distinct.

First,

$$p_{i+1}-p_{i-1}=e_{i-1}+e_i.$$

If $p_{i+1}=p_{i-1}$, then $e_i=-e_{i-1}$, requiring a turn of π, impossible since total turn is $<\pi/3$.

Next,

$$q_{i-1}=p_{i-1}+\omega e_{i-1}=p_i+(\omega-1)e_{i-1}=p_i+\omega^2e_{i-1},$$

so $q_{i-1}\ne p_i$. Also $q_i=p_i+\omega e_i\ne p_i$. Further,

$$q_{i-1}-p_{i-1}=\omega e_{i-1}\ne0,$$

and

$$q_i-p_{i+1}=(\omega-1)e_i=\omega^2e_i\ne0.$$

If $q_i=p_{i-1}$, then

$$p_i+\omega e_i=p_i-e_{i-1},$$

so $e_i=-\omega^{-1}e_{i-1}=e^{2\pi i/3}e_{i-1}$, a counterclockwise turn $2\pi/3$, impossible. If $q_{i-1}=p_{i+1}$, then $e_i=\omega^2e_{i-1}$, again a counterclockwise turn $2\pi/3$. If $q_{i-1}=q_i$, then $e_i=\omega e_{i-1}$, a counterclockwise turn $\pi/3$, impossible. Thus the four displayed neighbours are distinct. Since $\deg(p_i)=4$, they are exactly all neighbours of p_i.

For p_0, Swanepoel gives $\deg(p_0)=3$ and $p_1,q_0\in N(p_0)$. Let p_{-1} denote the third neighbour. For p_{11}, Swanepoel gives $\deg(p_{11})=4$ and $p_{10},q_{10}\in N(p_{11})$, leaving at most two further neighbours.

Lemma — Endpoint q-caps

$$N(q_1)\subseteq \{p_1,p_2,q_0,q_2,t_1\}\cup U_1,\qquad |U_1|\le 1.$$

Also,

$$N(q_9)\subseteq \{p_9,p_{10},q_8,q_{10},t_8\}\cup U_9,\qquad |U_9|\le 1.$$

Proof.

Swanepoel Lemma 8 applies to q_i for $1\le i\le 2m-5$. For $m=7$, this includes q_1 and q_9.

For q_1, the two extra neighbours outside $\{p_1,p_2,q_0,q_2\}$ include $t_1=s_1=r_2$. Therefore at most one further extra neighbour remains, giving the stated U_1.

For q_9, the two extra neighbours outside $\{p_9,p_{10},q_8,q_{10}\}$ include $t_8=s_8=r_9$. Therefore at most one further extra neighbour remains, giving the stated U_9.

Lemma — Boundary-boundary separation

If $b-a\ge 2$, then

$$|p_b-p_a|>1.$$

Proof.

If $b=a+2$, then

$$p_b-p_a=e_a+e_{a+1},$$

so

$$|p_b-p_a|=2\cos(\tau_{a+1}/2)>\sqrt{3}>1.$$

If $b-a\ge 3$, then

$$p_b-p_a=e_a+e_{a+1}+\cdots+e_{b-1}.$$

Project onto e_a. The first three terms contribute 1, $>1/2$, and $>1/2$, respectively. The projection is $>2>1$, so the length is >1.

Lemma — Boundary-to-q separation

If $a\notin\{b,b+1\}$, then

$$|p_a-q_b|>1.$$

Proof.

If $a<b$, then

$$q_b-p_a=e_a+\cdots+e_{b-1}+\omega e_b.$$

Project onto e_a. The e_a-term contributes 1, and the ωe_b-term contributes at least $1/2$. Thus the projection is >1.

If $a>b+1$, then

$$\begin{aligned}p_a-q_b&=e_b+\cdots+e_{a-1}-\omega e_b\\&=\bar\omega e_b+e_{b+1}+\cdots+e_{a-1}.\end{aligned}$$

For $a=b+2$, the vector is $\bar\omega e_b+e_{b+1}$, whose two summands have angle $\pi/3-\tau_{b+1}\in[0,\pi/3]$. Hence the squared length is at least 3. For $a\ge b+3$, project onto e_b: the first three terms contribute $1/2$, $>1/2$, and $>1/2$, so the projection is $>3/2>1$.

Lemma — Boundary-to-t separation

Let

$$t_b=p_b+\omega(e_b+e_{b+1}).$$

If p_a is not one of the explicitly displayed local neighbours of t_b, then

$$|p_a-t_b|>1.$$

Proof.

If $a<b$, then

$$t_b-p_a=e_a+\cdots+e_{b-1}+\omega e_b+\omega e_{b+1}.$$

Project onto e_a. The first e-term contributes 1, and the two ωe-terms each contribute at least $1/2$. Hence the projection is at least $2>1$.

If $a=b$, then

$$t_b-p_b=\omega(e_b+e_{b+1}),$$

whose length is $|e_b+e_{b+1}|>\sqrt{3}>1$.

If $a=b+1$, then

$$p_{b+1}-t_b=\bar\omega e_b-\omega e_{b+1}.$$

Writing $e_{b+1}=e_b e^{-i\tau_{b+1}}$, its squared length is

$$2-2\cos\left(\frac{2\pi}{3}-\tau_{b+1}\right)>1,$$

since $0\le\tau_{b+1}<\pi/3$.

If $a=b+2$, then

$$p_{b+2}-t_b=\bar\omega(e_b+e_{b+1}),$$

whose length is $>\sqrt{3}$.

If $a>b+2$, then

$$p_a-t_b=\bar\omega e_b+\bar\omega e_{b+1}+e_{b+2}+\cdots+e_{a-1}.$$

Project onto e_{b+2}. The terms e_{b+2}, $\bar\omega e_{b+1}$, and $\bar\omega e_b$ contribute at least 1, $1/2$, and $1/2$, respectively; all later terms have nonnegative projection. Hence the projection is at least $2>1$.

Lemma — q_7t_5 separation

If $t_4t_5\notin E(G)$, then

$$|q_7-t_5|>1.$$

Proof.

We prove $|q_7-t_5|\ge\sqrt{3}$. We have

$$q_7=p_7+\omega e_7,$$

$$t_5=p_5+\omega(e_5+e_6).$$

Therefore

$$\begin{aligned} q_7-t_5&=e_5+e_6+\omega e_7-\omega e_5-\omega e_6\\ &=\bar\omega(e_5+e_6)+\omega e_7. \end{aligned}$$

Rotate so $e_5=1$. Write

$$e_6=e^{-ia},$$

$$e_7=e^{-i(a+b)},$$

with $a,b\ge0$ and $a+b<\pi/3$. Multiplication by ω preserves length, so

$$|q_7-t_5|=\bigl|1+e^{-ia}+\omega^2e^{-i(a+b)}\bigr|.$$

Let

$$F(a,b)=1+e^{-ia}+\omega^2e^{-i(a+b)}.$$

Then

$$|F(a,b)|^2=3+2H(a,b),$$

where

$$H(a,b)=\cos a+\cos\left(\frac{2\pi}{3}-b\right)+\cos\left(\frac{2\pi}{3}-a-b\right).$$

For fixed a,

$$\frac{\partial H}{\partial b}=\sin\left(\frac{2\pi}{3}-b\right)+\sin\left(\frac{2\pi}{3}-a-b\right)\ge0,$$

because both arguments lie in $(\pi/3,2\pi/3]$. Hence $H(a,b)$ is minimized at $b=0$. Now

$$H(a,0)=\cos a-\frac12+\cos\left(\frac{2\pi}{3}-a\right)=\frac12\cos a-\frac12+\frac{\sqrt3}{2}\sin a.$$

Thus

$$2H(a,0)=\cos a+\sqrt3\sin a-1=2\cos(a-\pi/3)-1\ge 0,$$

since $0\le a<\pi/3$. Therefore $H(a,b)\ge 0$, and $|F(a,b)|^2\ge 3$.

Cap lemmas

Lemma — Swanepoel t-caps

$$|N(t_5)\setminus\{t_4,t_6,q_5,q_6\}|\le 2,$$

$$|N(t_6)\setminus\{t_5,t_7,q_6,q_7\}|\le 2,$$

$$|N(t_7)\setminus\{t_6,t_8,q_7,q_8\}|\le 2.$$

Proof.

Apply the Swanepoel Lemma 9 cap from Proposition prop:SwanepoelInput to the equality triples $(4,5,6)$, $(5,6,7)$, and $(6,7,8)$. For $t_5=s_5$, the cap set is $\{s_4,s_6,q_5,q_6\}=\{t_4,t_6,q_5,q_6\}$. The other two cases are identical.

Definition — Port and saturation

In the fully flat branch, the t_i-port, for $i=5,6,7$, is the closed angular interval of possible extra neighbours of t_i between the endpoint directions 60° and 120° relative to the flat strip direction. It is saturated if both endpoint positions

$$x_i=t_i+\omega,$$

$$y_i=t_i+\omega^2$$

are occupied by vertices of the point set.

Lemma — Fully flat port cap

In the fully flat branch, every extra neighbour of t_i, $i=5,6,7$, outside the strip-neighbour set lies in the t_i-port. If the port is not saturated, it contributes at most one extra neighbour. If one endpoint is already present and the port is not saturated, it contributes no further extra neighbour. Consecutive saturated ports share one endpoint.

Proof.

Place $t_i=(0,0)$ with the flat strip direction horizontal. The four forced strip-neighbour directions at t_i are

$$0^\circ,\quad 180^\circ,\quad 240^\circ,\quad 300^\circ.$$

A further unit neighbour must have angular separation at least 60° from each. These four neighbours forbid the open arcs

$$(-60^\circ,60^\circ),$$

$$(120^\circ,240^\circ),$$

$$(180^\circ,300^\circ),$$

and

$$(240^\circ,360^\circ).$$

The only remaining arc is $[60^\circ,120^\circ]$.

If two extra neighbours lie in this closed 60° interval, their angular separation is at most 60°. Since their mutual distance must be at least 1, the angular separation is exactly 60°; hence they are the two endpoints. Thus a non-saturated port contributes at most one extra neighbour.

If one endpoint is present and another extra neighbour existed, it would either be an interior point, which is too close to the endpoint, or the other endpoint, which would saturate the port. Hence no further extra neighbour exists in the non-saturated case.

The sharing relation follows from

$$t_i+\left(\frac12,\frac{\sqrt3}{2}\right) =t_{i+1}+\left(-\frac12,\frac{\sqrt3}{2}\right).$$

Lemma — Branch-qualified degree caps

In the fully flat branch, if the t_7-port is saturated, then

$$N(t_8)\subseteq\{q_8,q_9,t_7,x_7\}\cup U_8, \qquad |U_8|\le 2.$$

In the all-saturated branch,

$$N(x_6)\subseteq\{t_6,t_7,x_5,x_7\}\cup V_6, \qquad |V_6|\le 2.$$

If t_6,t_7 are saturated but t_5 is not saturated, then

$$N(x_6)\subseteq\{t_6,t_7,y_6,x_7\}\cup V_6, \qquad |V_6|\le 2.$$

Proof.

In fully flat coordinates,

$$t_8=3+2\omega.$$

If t_7 is saturated, then

$$x_7=2+3\omega.$$

The vertices

$$q_8=3+\omega,$$

$$q_9=4+\omega,$$

$$t_7=2+2\omega,$$

$$x_7=2+3\omega$$

are distinct and have differences from t_8 equal to

$$-\omega,$$

$$1-\omega,$$

$$-1,$$

$$-1+\omega,$$

all of length 1. Since $\deg(t_8)\le 6$, at most two further neighbours exist.

For $x_6=1+3\omega$ in the all-saturated branch, the vertices

$$t_6=1+2\omega,$$

$$t_7=2+2\omega,$$

$$x_5=3\omega,$$

$$x_7=2+3\omega$$

are distinct and have differences from x_6 equal to

$$-\omega,$$

$$1-\omega,$$

$$-1,$$

$$1.$$

All have length 1, so at most two further neighbours exist.

If t_6,t_7 are saturated but t_5 is not, then the left endpoint adjacent to x_6 is the actual vertex $y_6=3\omega$, not an actual saturated-port vertex x_5. The four known neighbours are

$$t_6,t_7,y_6,x_7,$$

again with differences $-\omega$, $1-\omega$, -1, and 1 from x_6. The degree-six bound gives the cap.

Lemma — Neighbourhood inclusion

Let S be independent. Suppose

$$N(v)\subseteq A\cup C_1\cup\cdots\cup C_r\cup S$$

for every $v\in S$. Then

$$N(S)\subseteq A\cup C_1\cup\cdots\cup C_r.$$

Thus

$$|N(S)|\le |A|+\sum_{j=1}^r |C_j|.$$

Proof.

If $x\in N(S)$, then $x\notin S$ and $x\in N(v)$ for some $v\in S$. By the assumed containment, $x\in A\cup C_1\cup\cdots\cup C_r\cup S$. Since $x\notin S$, $x\in A\cup C_1\cup\cdots\cup C_r$.

The t_4t_5-absent branch

Lemma

If $t_4t_5\notin E(G)$, then

$$S_0=\{p_0,p_2,p_4,p_6,p_9,q_7,t_5\}$$

is independent and

$$|N(S_0)|\le 20.$$

Proof.

The selected boundary vertices are mutually nonadjacent by Lemma lem:bb. They are nonadjacent to q_7 by Lemma lem:bq and to t_5 by Lemma lem:bt. Finally, q_7t_5 is absent by Lemma lem:q7t5. Thus S_0 is independent.

The selected boundary vertices contribute only

$$p_{-1},p_1,p_3,p_5,p_7,p_8,p_{10},\ q_0,q_1,q_2,q_3,q_4,q_5,q_6,q_8,q_9.$$

The vertex q_7 contributes only

$$p_7,p_8,q_6,q_8,t_6,t_7.$$

The vertex t_5 has displayed possible neighbours t_4,t_6,q_5,q_6. Since t_4t_5 is absent, the displayed neighbours reduce to t_6,q_5,q_6, plus a Swanepoel cap C_5 with $|C_5|\le 2$.

Let

$$\begin{aligned}
A_0=\{&p_{-1},p_1,p_3,p_5,p_7,p_8,p_{10},q_0,q_1,q_2,q_3,q_4,q_5,q_6,q_8,q_9,t_6,t_7\}.
\end{aligned}$$

Then $|A_0|\le 18$ and

$$N(S_0)\subseteq A_0\cup C_5.$$

Therefore

$$|N(S_0)|\le 18+2=20.$$

The t_4t_5-present branches before full flatness

Assume throughout this section that $t_4t_5\in E(G)$. By Corollary cor:t4t5,

$$\tau_5=\tau_6=0.$$

Lemma

If $t_4t_5\in E(G)$ and $\tau_7>0$, then

$$S_1=\{p_6,p_8,q_4,t_5,t_6,t_7\}$$

is independent and

$$|N(S_1)|\le 17.$$

Proof.

The boundary pair p_6,p_8 is nonadjacent by Lemma lem:bb. The vertices p_6,p_8 are nonadjacent to q_4 by Lemma lem:bq and to t_5,t_6,t_7 by Lemma lem:bt.

The unique break inequality gives

$$\tau_3+2\tau_4+\tau_5\ge\frac{\pi}{3}.$$

Since $\tau_5=0$ and the total turn is $<\pi/3$, we get $\tau_4>0$. By the general q-turn criterion, Lemma lem:q-turn,

$$q_3q_4\notin E(G).$$

Swanepoel Lemma 8 gives

$$N(q_4)\subseteq\{p_4,p_5,q_3,q_5,r_4,s_4\}.$$

Since q_3q_4 is absent and $s_4=t_4$,

$$N(q_4)\subseteq\{p_4,p_5,q_5,r_4,t_4\}.$$

Normalize $p_4=0$ and $e_4=e_5=e_6=1$. Let $a=\tau_7>0$ and $b=\tau_8\ge0$. Then $a+b<\pi/3$ and

$$e_7=e^{-ia},$$

$$e_8=e^{-i(a+b)}.$$

Thus

$$q_4=\omega,$$

$$t_5=1+2\omega,$$

$$t_6=2+\omega(1+e^{-ia}),$$

$$t_7=3+\omega(e^{-ia}+e^{-i(a+b)}).$$

Now

$$t_5-q_4=1+\omega,\qquad |t_5-q_4|=\sqrt3>1.$$

Also,

$$t_6-q_4=2+\omega e^{-ia},$$

whose real part is $2+\cos(\pi/3-a)>1$. Similarly,

$$t_7-q_4=3-\omega+\omega e^{-ia}+\omega e^{-i(a+b)},$$

whose real part is

$$3-\frac12+\cos(\pi/3-a)+\cos(\pi/3-a-b)>1.$$

By Lemma lem:t-equality, t_5t_6 is absent because $\tau_6+\tau_7>0$, and t_6t_7 is absent because $\tau_7+\tau_8>0$. Finally,

$$t_7-t_5=2\bar\omega+\omega(e^{-ia}+e^{-i(a+b)}).$$

Projection in the e_5 direction gives

$$2\cos(\pi/3)+\cos(\pi/3-a)+\cos(\pi/3-a-b)>2,$$

so $|t_7-t_5|>1$. Thus S_1 is independent.

Let

$$A_1=\{p_4,p_5,p_7,p_9,q_5,q_6,q_7,q_8,r_4,t_4,t_8\}.$$

Then $|A_1|\le 11$. The non-capped neighbours of the selected vertices lie in A_1. The vertices t_5,t_6,t_7 have Swanepoel caps C_5,C_6,C_7, each of size at most 2. Therefore

$$N(S_1)\subseteq A_1\cup C_5\cup C_6\cup C_7,$$

and

$$|N(S_1)|\le 11+2+2+2=17.$$

Lemma

If $t_4t_5\in E(G)$, $\tau_7=0$, and $\tau_8+\tau_9>0$, then

$$S_2=\{p_0,p_3,p_5,p_8,q_1,q_6,t_7\}$$

is independent and

$$|N(S_2)|\le 20.$$

Proof.

The selected boundary vertices are mutually nonadjacent by Lemma lem:bb. They are nonadjacent to q_1,q_6 by Lemma lem:bq and to t_7 by Lemma lem:bt.

Rotate so $e_1=1$. Then

$$q_6-q_1=e_1+e_2+e_3+e_4+e_5+\omega e_6-\omega e_1.$$

By Lemma lem:cone,

$$\mathrm{Re}(q_6-q_1)\ge 5\cdot\frac12+\frac12-\frac12=\frac52>1.$$

Thus q_1q_6 is absent.

Next,

$$t_7-q_1=e_1+e_2+e_3+e_4+e_5+e_6+\omega e_7+\omega e_8-\omega e_1.$$

Thus

$$\mathrm{Re}(t_7-q_1)\ge 6\cdot\frac12+\frac12+\frac12-\frac12=\frac72>1.$$

So q_1t_7 is absent.

Since $\tau_6=\tau_7=0$, we have $e_6=e_7$, and

$$t_7-q_6=e_6+\omega e_8.$$

The angle between e_6 and ωe_8 is $\pi/3-\tau_8$. Hence

$$|t_7-q_6|^2=2+2\cos(\pi/3-\tau_8)\ge 3>1.$$

Thus q_6t_7 is absent, and S_2 is independent.

By Lemma lem:q-caps,

$$N(q_1)\subseteq\{p_1,p_2,q_0,q_2,t_1\}\cup U_1,\qquad |U_1|\le 1.$$

Also,

$$N(q_6)\subseteq\{p_6,p_7,q_5,q_7,t_5,t_6\}.$$

Since $\tau_8+\tau_9>0$, Lemma lem:t-equality gives $t_7t_8\notin E(G)$. The Swanepoel cap gives

$$N(t_7)\subseteq\{q_7,q_8,t_6\}\cup C_7,\qquad |C_7|\le 2.$$

Let

$$\begin{aligned}
A_2=\{&p_{-1},p_1,p_2,p_4,p_6,p_7,p_9,q_0,q_2,q_3,q_4,q_5,q_7,q_8,t_1,t_5,t_6\}.
\end{aligned}$$

Then $|A_2|\le 17$ and

$$N(S_2)\subseteq A_2\cup U_1\cup C_7.$$

Therefore

$$|N(S_2)|\le 17+1+2=20.$$

Fully flat branch

Assume

$$\tau_7=\tau_8=\tau_9=0.$$

Normalize

$$p_5=0,$$

$$e_5=e_6=e_7=e_8=e_9=1.$$

Then

$$p_6=1,\quad p_7=2,\quad p_8=3,\quad p_9=4,\quad p_{10}=5,$$

$$q_5=\omega,\quad q_6=1+\omega,\quad q_7=2+\omega,\quad q_8=3+\omega,\quad q_9=4+\omega,$$

and

$$t_5=2\omega,\quad t_6=1+2\omega,\quad t_7=2+2\omega,\quad t_8=3+2\omega.$$

Also

$$p_{11}=5+e^{-i\theta},\quad 0\le\theta<\frac{\pi}{3}.$$

When a port is saturated,

$$x_i=t_i+\omega,\quad y_i=t_i+\omega^2.$$

Thus

$$x_5=3\omega,\quad x_6=1+3\omega,\quad x_7=2+3\omega,$$

and

$$y_6=x_5,\quad y_7=x_6.$$

Lemma

If t_5 and t_7 are saturated, then t_6 is saturated. Hence the possible non-all-saturated patterns are exactly

$$\varnothing,\quad t_5,\quad t_6,\quad t_7,\quad t_5t_6,\quad t_6t_7.$$

Proof.

Saturation at t_5 supplies the endpoint $x_5=y_6$ of the t_6-port. Saturation at t_7 supplies the endpoint $x_6=y_7$ of the t_6-port. Thus both endpoints of the t_6-port are present, so t_6 is saturated.

Lemma — No saturated port

If none of the ports at t_5,t_6,t_7 is saturated, then the set

$$S_\varnothing=\{p_6,p_8,p_{11},q_9,t_5,t_7\}$$

is independent and satisfies $|N(S_\varnothing)|\le 17$.

Proof.

The non-p_{11} selected-pair differences are

$$p_8-p_6=2,$$

$$q_9-p_6=3+\omega,$$

$$t_5-p_6=2\omega-1,$$

$$t_7-p_6=1+2\omega,$$

$$q_9-p_8=1+\omega,$$

t_5-p_8=2ω-3,

t_7-p_8=2ω-1,

t_5-q_9=ω-4,

t_7-q_9=ω-2,

and

t_7-t_5=2.

None is 0 or one of the six triangular-lattice unit vectors. Since the graph has minimum distance 1, these differences all have length >1, except where the displayed length is visibly 2 or √3.

For p_11,

p_11-q_9=1+e^-itheta-ω,

with real part 1+costheta-1/2>1. Also,

p_11-t_5=5+e^-itheta-2ω,

with real part 4+costheta>1,

p_11-t_7=3+e^-itheta-2ω,

with real part 2+costheta>1,

p_11-p_6=4+e^-itheta,

with real part 4+costheta>1, and

p_11-p_8=2+e^-itheta,

with real part 2+costheta>1. Thus S_varnothing is independent.

Let

A_varnothing=\p_5,p_7,p_9,p_10,q_5,q_6,q_7,q_8,q_10,t_4,t_6,t_8\.

Then |A_varnothing|≤12 and

N(S_varnothing)⊆ A_varnothing∪ U_11∪ U_9∪ C_5∪ C_7,

where |U_11|≤2, |U_9|≤1, and |C_5|,|C_7|≤1 are the flat-port caps for the unsaturated t_5 and t_7 ports. Hence

|N(S_varnothing)|≤12+2+1+1+1=17.

Lemma — Only t_5 saturated

If only the t_5-port is saturated, then

S_5=\p_5,p_8,p_11,q_6,q_9,t_7\

is independent and satisfies |N(S_5)|≤17.

Proof.

The selected-pair differences not involving p_11 are

p_5-p_8=-3,

p_5-q_6=-(1+ω),

p_5-q_9=-(4+ω),

p_5-t_7=-(2+2ω),

p_8-q_6=2-ω,

p_8-q_9=-(1+ω),

p_8-t_7=1-2ω,

q_6-q_9=-3,

q_6-t_7=-(1+ω),

and

q_9-t_7=2-ω.

None is 0 or a triangular-lattice unit vector, and the cases of lengths 3 or √3 are immediate.

For p_11,

p_11-p_5=5+e^-itheta,

whose real part is $5+\cos\theta>1$;

$$p_{11}-p_8=2+e^{-i\theta},$$

whose real part is $2+\cos\theta>1$;

$$p_{11}-q_6=4+e^{-i\theta}-\omega,$$

whose real part is $4+\cos\theta-1/2>1$;

$$p_{11}-q_9=1+e^{-i\theta}-\omega,$$

whose real part is $1+\cos\theta-1/2>1$; and

$$p_{11}-t_7=3+e^{-i\theta}-2\omega,$$

whose real part is $2+\cos\theta>1$. Thus S_5 is independent.

Let

$$A_5=\{p_4,p_6,p_7,p_9,p_{10},q_4,q_5,q_7,q_8,q_{10},t_5,t_6,t_8\}.$$

Then $|A_5|\le 13$ and

$$N(S_5)\subseteq A_5\cup U_{11}\cup U_9\cup C_7,$$

where $|U_{11}|\le 2$, $|U_9|\le 1$, and $|C_7|\le 1$. Hence

$$|N(S_5)|\le 13+2+1+1=17.$$

Lemma — Only t_6 saturated

If only the t_6-port is saturated, then

$$S_6=\{p_5,p_8,p_{11},q_6,q_9,t_7\}$$

is independent and satisfies $|N(S_6)|\le 17$.

Proof.

This is the same selected set as in Lemma lem:row5, so it is independent. Let

$$A_6=\{p_4,p_6,p_7,p_9,p_{10},q_4,q_5,q_7,q_8,q_{10},t_5,t_6,t_8,x_6\}.$$

Then $|A_6|\le 14$. The t_7-port is not saturated, but the saturated t_6-port supplies one endpoint $x_6=y_7$ of the t_7-port. By Lemma lem:port, t_7 has no further port neighbour. Therefore

$$N(S_6)\subseteq A_6\cup U_{11}\cup U_9,$$

and

$$|N(S_6)|\le 14+2+1=17.$$

Lemma — Only t_7 saturated

If only the t_7-port is saturated, then

$$S_7=\{p_0,p_3,p_6,q_1,q_4,t_5\}$$

is independent and satisfies $|N(S_7)|\le 17$.

Proof.

Boundary-boundary nonadjacency follows from Lemma lem:bb; boundary-to-q and boundary-to-t nonadjacency follow from Lemmas lem:bq and lem:bt.

The remaining auxiliary pairs are q_1q_4, q_1t_5, q_4t_5. First,

$$q_4-q_1=e_1+e_2+e_3+\omega e_4-\omega e_1,$$

so

$$\mathrm{Re}(q_4-q_1)\ge \frac{3}{2}>1.$$

Thus q_1q_4 is absent. Next,

$$t_5-q_1=e_1+e_2+e_3+e_4+\omega e_5+\omega e_6-\omega e_1,$$

so

$$\mathrm{Re}(t_5-q_1)\ge \frac{5}{2}>1.$$

Thus q_1t_5 is absent. Finally, since $e_4=e_5=e_6$, normalizing $p_4=0$ gives

$$q_4=\omega,$$

$$t_5=1+2\omega.$$

Thus

$$t_5-q_4=1+\omega,$$

and the distance is $\sqrt{3}>1$. Hence S_7 is independent.

Let

$$A_7=\{p_{-1},p_1,p_2,p_4,p_5,p_7,q_0,q_2,q_3,q_5,q_6,r_4,t_1,t_4,t_6\}.$$

Then $|A_7|\le 15$ and

$$N(S_7)\subseteq A_7\cup U_1\cup C_5,$$

with $|U_1|\le 1$ and $|C_5|\le 1$. Therefore

$$|N(S_7)|\le 15+1+1=17.$$

Lemma — t_5,t_6 saturated

If the t_5- and t_6-ports are saturated and the t_7-port is not saturated, then

$$S_{56}=\{p_5,p_8,p_{11},q_6,q_9,t_7\}$$

is independent and satisfies $|N(S_{56})|\le 17$.

Proof.

The selected set is the same as in Lemma lem:row5, so it is independent. Let

$$A_{56}=\{p_4,p_6,p_7,p_9,p_{10},q_4,q_5,q_7,q_8,q_{10},t_5,t_6,t_8,x_6\}.$$

Then $|A_{56}|\le 14$. The t_7-port is not saturated, and the saturated t_6-port supplies the endpoint $x_6=y_7$. By Lemma lem:port, t_7 has no further port neighbour. Hence

$$N(S_{56})\subseteq A_{56}\cup U_{11}\cup U_9,$$

and

$$|N(S_{56})|\le 14+2+1=17.$$

Lemma — t_6,t_7 saturated

If the t_6- and t_7-ports are saturated and the t_5-port is not saturated, then

$$S_{67}=\{p_6,p_9,q_7,t_5,t_8,x_6\}$$

is independent and satisfies $|N(S_{67})|\le 17$.

Proof.

In fully flat coordinates,

$$p_6=1,\quad p_9=4,\quad q_7=2+\omega,\quad t_5=2\omega,\quad t_8=3+2\omega,\quad x_6=1+3\omega.$$

The selected-pair differences are

$$p_9-p_6=3,$$

$$q_7-p_6=1+\omega,$$

$$t_5-p_6=2\omega-1,$$

$$t_8-p_6=2+2\omega,$$

$$x_6-p_6=3\omega,$$

$$q_7-p_9=-2+\omega,$$

$$t_5-p_9=2\omega-4,$$

$$t_8-p_9=-1+2\omega,$$

$$x_6-p_9=-3+3\omega,$$

$$t_5-q_7=\omega-2,$$

$$t_8-q_7=1+\omega,$$

$$x_6-q_7=-1+2\omega,$$

$$t_8-t_5=3,$$

$$x_6-t_5=1+\omega,$$

$$x_6-t_8=-2+\omega.$$

None is 0 or one of $\pm1,\pm\omega,\pm\omega^2$. Thus no selected distance is 1, and all selected distances are >1. Hence S_{67} is independent.

Let

$$A_{67}=\{p_5,p_7,p_8,p_{10},q_5,q_6,q_8,q_9,t_4,t_6,t_7,y_6,x_7\}.$$

Then $|A_{67}|\le 13$. Since t_7 is saturated, Lemma lem:degree-caps gives

$$N(t_8)\subseteq\{q_8,q_9,t_7,x_7\}\cup U_8, \qquad |U_8|\le 2.$$

Since t_6,t_7 are saturated but t_5 is not, Lemma lem:degree-caps gives

$$N(x_6)\subseteq\{t_6,t_7,y_6,x_7\}\cup V_6, \qquad |V_6|\le 2.$$

Finally, $y_6=x_5$ occupies one endpoint of the t_5-port, and t_5 is not saturated, so Lemma lem:port gives no further t_5-port neighbour. Therefore

$$N(S_{67})\subseteq A_{67}\cup U_8\cup V_6,$$

and

$$|N(S_{67})|\le 13+2+2=17.$$

Corollary

In the fully flat branch, if the ports at t_5,t_6,t_7 are not all saturated, then there is an independent set S with

$$|S|=6,$$

$$|N(S)|\le 17.$$

Proof.

By Lemma lem:flat-patterns, the non-all-saturated patterns are exactly the six patterns in Lemmas lem:row0--lem:row67. Each lemma supplies the required set.

The all-saturated branch

Assume the ports at t_5,t_6,t_7 are all saturated. Let

$$S_3=\{p_0,p_3,p_6,p_9,q_1,q_4,q_7,t_5,t_8,x_6\}.$$

Lemma

The set S_3 is independent and satisfies

$$|N(S_3)|\le 28.$$

Proof.

Boundary-boundary nonadjacency follows from Lemma lem:bb; boundary-to-q and boundary-to-t nonadjacency follow from Lemmas lem:bq and lem:bt.

We check x_6 against selected boundary vertices. In the fully flat branch,

$$x_6=p_6+3\omega e_6.$$

Thus

$$|x_6-p_6|=3.$$

In flat coordinates,

$$x_6-p_9=-3+3\omega,$$

so $|x_6-p_9|=3$. For p_3,

$$x_6-p_3=e_3+e_4+e_5+3\omega e_6.$$

Rotating so $e_3=1$ gives

$$\mathrm{Re}(x_6-p_3)>1+\frac12+\frac12+3\cdot\frac12=\frac72>1.$$

For p_0,

$$x_6-p_0=e_0+e_1+e_2+e_3+e_4+e_5+3\omega e_6.$$

Rotating so $e_0=1$ gives

$$\mathrm{Re}(x_6-p_0)>1+5\cdot\frac12+3\cdot\frac12=5>1.$$

Thus x_6 is not adjacent to the selected boundary vertices.

Now consider pairs involving q_1:

$$q_4-q_1=e_1+e_2+e_3+\omega e_4-\omega e_1,$$

so $\mathrm{Re}(q_4-q_1)\ge3/2>1$. Also,

$$q_7-q_1=e_1+\cdots+e_6+\omega e_7-\omega e_1,$$

so $\mathrm{Re}(q_7-q_1)\ge3>1$. Further,

$$t_5-q_1=e_1+\cdots+e_4+\omega e_5+\omega e_6-\omega e_1,$$

so $\mathrm{Re}(t_5-q_1)\ge5/2>1$. Also,

$$t_8-q_1=e_1+\cdots+e_7+\omega e_8+\omega e_9-\omega e_1,$$

so $\mathrm{Re}(t_8-q_1)\ge7/2>1$. Finally,

$$x_6-q_1=e_1+\cdots+e_5+3\omega e_6-\omega e_1,$$

so $\mathrm{Re}(x_6-q_1)\ge7/2>1$.

Now check pairs involving q_4. The break inequality gives

$$\tau_3+2\tau_4+\tau_5\ge\frac{\pi}{3}.$$

Since $\tau_5=0$ and total turn is $<\pi/3$, we get $\tau_4>0$. By the general q-turn criterion, $q_3q_4\notin E(G)$. In the fully flat normalization,

$$p_4=-1,\quad q_4=-1+\omega.$$

Also,

$$q_7=2+\omega,\quad t_5=2\omega,\quad t_8=3+2\omega,\quad x_6=1+3\omega.$$

Thus

$$q_7-q_4=3,$$

$$t_5-q_4=1+\omega,$$

$$t_8-q_4=4+\omega,$$

$$x_6-q_4=2+2\omega.$$

All have length greater than 1.

Among q_7,t_5,t_8,x_6, the differences are

$$t_5-q_7=\omega-2,$$

$$t_8-q_7=1+\omega,$$

$$x_6-q_7=-1+2\omega,$$

$$t_8-t_5=3,$$

$$x_6-t_5=1+\omega,$$

$$x_6-t_8=-2+\omega.$$

None is a triangular-lattice unit vector, so every selected-pair distance is >1. Hence S_3 is independent.

Now define

$$\begin{aligned}
A_3=\{&p_{-1},p_1,p_2,p_4,p_5,p_7,p_8,p_{10},q_0,q_2,q_3,q_5,q_6,q_8,q_9,\\
&r_4,t_1,t_4,t_6,t_7,x_5,x_7,y_5\}.
\end{aligned}$$

Then $|A_3|\le23$.

The local neighbourhood inclusions are

$$N(p_0),N(p_3),N(p_6),N(p_9)\subseteq A_3,$$

$$N(q_1)\subseteq A_3\cup U_1,$$

$$|U_1|\le 1.$$

Since q_3q_4 is absent and $s_4=t_4$,

$$N(q_4)\subseteq A_3.$$

Also,

$$N(q_7)\subseteq A_3,$$
$$N(t_5)\subseteq A_3.$$

Since all ports are saturated, Lemma lem:degree-caps gives

$$N(t_8)\subseteq A_3\cup U_8,$$
$$|U_8|\le 2,$$

and

$$N(x_6)\subseteq A_3\cup V_6,$$
$$|V_6|\le 2.$$

Thus

$$N(S_3)\subseteq A_3\cup U_1\cup U_8\cup V_6.$$

Consequently,

$$|N(S_3)|\le 23+1+2+2=28.$$

Final proof

Theorem

Relative to Swanepoel's published broken-lattice inputs stated in Proposition prop:SwanepoelInput, every Euclidean minimum-distance graph G satisfies

$$\alpha(G)\ge \frac{7}{27}|V(G)|.$$

Consequently,

$$g(n)\ge \lceil \frac{7n}{27}\rceil.$$

Proof.

Assume that G is a smallest counterexample. By Lemma lem:deletion, every independent set S in G must satisfy

$$|N(S)|>\lfloor \frac{20|S|}{7}\rfloor.$$

By Lemma lem:unique-break, the only possible equality break is $s_3\ne r_4$, and t_i is defined for $i=1,2,4,5,6,7,8$.

If $t_4t_5\notin E(G)$, Lemma lem:S0 gives an independent set S_0 with $|S_0|=7$ and

$$|N(S_0)|\le 20=\lfloor \frac{20\cdot 7}{7}\rfloor,$$

contradiction. Hence $t_4t_5\in E(G)$.

By Corollary cor:t4t5, $\tau_5=\tau_6=0$. If $\tau_7>0$, Lemma lem:S1 gives an independent set S_1 with $|S_1|=6$ and

$$|N(S_1)|\le 17=\lfloor \frac{20\cdot 6}{7}\rfloor,$$

contradiction.

If $\tau_7=0$ and $\tau_8+\tau_9>0$, Lemma lem:S2 gives an independent set S_2 with $|S_2|=7$ and

$$|N(S_2)|\le 20=\lfloor \frac{20\cdot 7}{7}\rfloor,$$

contradiction.

Therefore $\tau_7=\tau_8=\tau_9=0$. If the ports at t_5,t_6,t_7 are not all saturated, Corollary cor:flat-nonall gives an independent set S with $|S|=6$ and

$$|N(S)|\le 17=\lfloor \frac{20\cdot 6}{7}\rfloor,$$

contradiction.

Thus all three ports are saturated. Lemma lem:S3 gives an independent set S_3 with $|S_3|=10$ and

$$|N(S_3)|\le 28=\lfloor \frac{20\cdot 10}{7}\rfloor,$$

contradiction.

Every branch contradicts the deletion criterion. Therefore no smallest counterexample exists. Hence

$$\alpha(G)\ge\frac{7}{27}|V(G)|.$$

Since $\alpha(G)$ is an integer,

$$\alpha(G)\ge\left\lceil\frac{7|V(G)|}{27}\right\rceil.$$

Taking the minimum over all n-vertex Euclidean minimum-distance graphs gives

$$g(n)\ge \left\lceil\frac{7n}{27}\right\rceil.$$

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